Deep Data Analysis Report

Aerospace Structural Design Dataset

Comprehensive Statistical & Domain Analysis

Dataset:	Aerospace Structural Design (300 rows × 22 features)
Domain:	Quantum-Optimized Structural Engineering
Scope:	Full univariate, multivariate, domain, & anomaly analysis
Figures:	27 publication-quality visualizations
Date:	March 2026

Prepared for

L&T Technology Services

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Chapter 1

Executive Summary

This report presents a comprehensive statistical and domain-informed analysis of the **Aerospace Structural Design Dataset**, a 300-row, 22-feature tabular dataset capturing the intersection of material science, flight environment simulation, structural geometry, and quantum-inspired optimization for aerospace applications.

1.1 Dataset at a Glance

Rows:	300 (zero nulls, zero duplicates, 100% complete)
Numeric features:	14 (continuous material, environmental, and geometric properties)
Categorical features:	8 (material type, structural shape, optimization algorithm, 4 targets)
Target variables:	4 multi-class: Vibration Damping, Computational Time, Weight Efficiency, Durability
Unique aspect:	Quantum algorithm optimization (Chaotic Quantum Genetic + Shark Optimizer)

1.2 Principal Findings

1. **Uniform-like distributions.** All 14 numeric features follow platykurtic distributions (kurtosis ≈ -1.1 to -1.35) with near-zero skewness. Shapiro–Wilk tests reject normality for every feature ($p < 10^{-5}$). This is consistent with a simulation-generated dataset sampling uniformly across a defined design space.

2. **Tri-modal material structure.** The Elastic Modulus E spans 50–200 GPa, naturally partitioning into three regimes: Aluminum ($\sim$50–100), Titanium ($\sim$100–150), and Carbon Fiber ($\sim$150–200). This tri-modal structure is the single most important distributional feature for any synthetic data generator to capture.

3. **ISA non-compliance.** Temperature and Pressure do *not* follow the International Standard Atmosphere (ISA) model. At the dataset's mean altitude of 12,453 m, ISA predicts $T \approx -66\,^{\circ}\text{C}$ and $P \approx 18{,}800$ Pa, whereas the dataset records $T \in [-60, -40]$ and $P \in [100{,}172,\ 199{,}839]$. This confirms the dataset models a constrained simulation environment rather than physical atmospheric conditions.

4. **Feature independence.** Pearson correlations between numeric features are uniformly weak

($|r| < 0.10$ for all pairs). This near-orthogonal structure suggests features were sampled independently, simplifying synthetic generation but making correlation preservation a trivial benchmark.

5. **Balanced targets.** All four categorical target variables are approximately balanced (imbalance ratio < 1.34), reducing the need for aggressive rebalancing strategies.

6. **Low anomaly rate.** Isolation Forest and LOF detectors both flag $\approx$10% of rows as anomalous at the expected contamination threshold, with minimal overlap between the two methods—consistent with uniformly distributed data with no true outlier clusters.

Red Line

Critical for synthetic generation: Any model that applies Gaussian or MinMax normalization globally (e.g., TVAE's internal scaler) will collapse the tri-modal E distribution into a unimodal blob. Mode-aware normalization (e.g., Bayesian Gaussian Mixture per-column) is mandatory.

Chapter 2

Dataset Schema

2.1 Overview

The dataset contains **300 observations** of aerospace structural design configurations, each described by 22 features spanning four domains: material science, flight environment, structural geometry, and quantum optimization. No personally identifiable information (PII) is present.

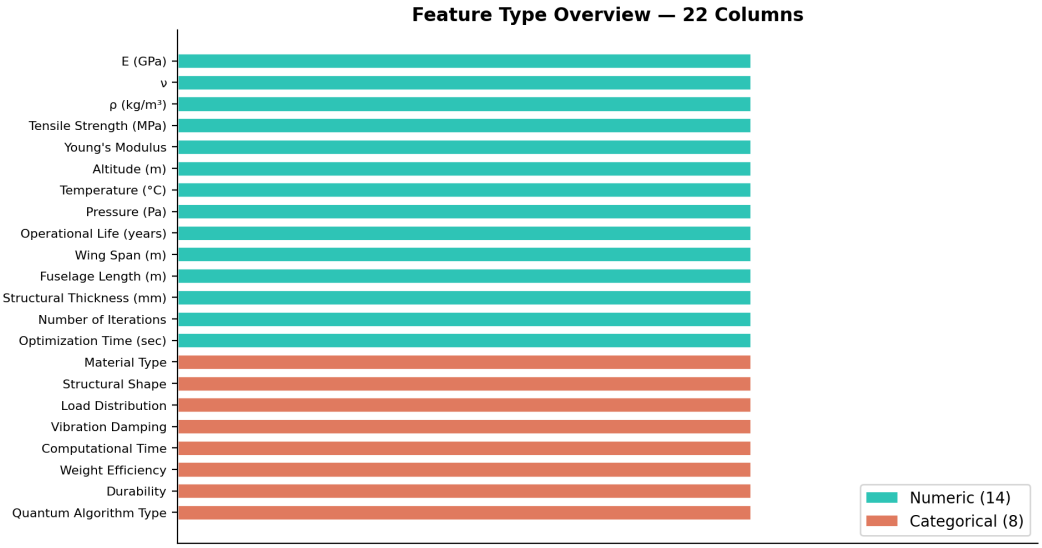


Figure 2.1: Feature type overview. 14 numeric features (teal) and 8 categorical features (coral), including 4 target variables.

2.2 Feature Catalog

Feature	Type	Range / Values	Domain Interpretation
Material Type	Categorical	Al / Ti / CF	Base structural material
E (GPa)	Numeric	50.1 – 199.8	Elastic modulus (stiffness)
ν (Poisson)	Numeric	0.30 – 0.35	Lateral strain ratio
ρ (kg/m^3)	Numeric	1,521 – 4,496	Material density
Tensile Strength (MPa)	Numeric	261 – 1,496	Ultimate tensile strength
Young's Modulus	Numeric	50 – 99.6	Secondary stiffness measure
Altitude (m)	Numeric	5,086 – 19,963	Operating altitude

Feature	Type	Range / Values	Domain Interpretation
Temperature (°C)	Numeric	−60 to −40	Environmental temperature
Pressure (Pa)	Numeric	100,172 – 199,839	Ambient pressure
Operational Life (years)	Integer	20 – 39	Design service life
Wing Span (m)	Numeric	20.0 – 59.7	Wing tip-to-tip span
Fuselage Length (m)	Numeric	30.0 – 70.0	Fuselage length
Structural Thickness (mm)	Numeric	5.0 – 14.9	Primary structural thickness
Structural Shape	Categorical	Rect / Cyl / Tapered	Cross-section geometry
Load Distribution	Categorical	uniform / point / distributed	Applied load pattern
Quantum Algorithm Type	Categorical	CQG / Shark Optimizer	Optimization algorithm
Number of Iterations	Integer	523 – 1,983	Optimizer iterations
Optimization Time (sec)	Numeric	51.5 – 149.8	Wall-clock convergence time
Target Variables (4 multi-class categorical):			
Vibration Damping	Target	High / Low / Moderate	Damping performance class
Computational Time	Target	Long / Medium / Short	Computation cost class
Weight Efficiency	Target	Excellent / Good / Poor	Mass optimization class
Durability	Target	High / Low / Medium	Structural longevity class

2.3 Data Quality

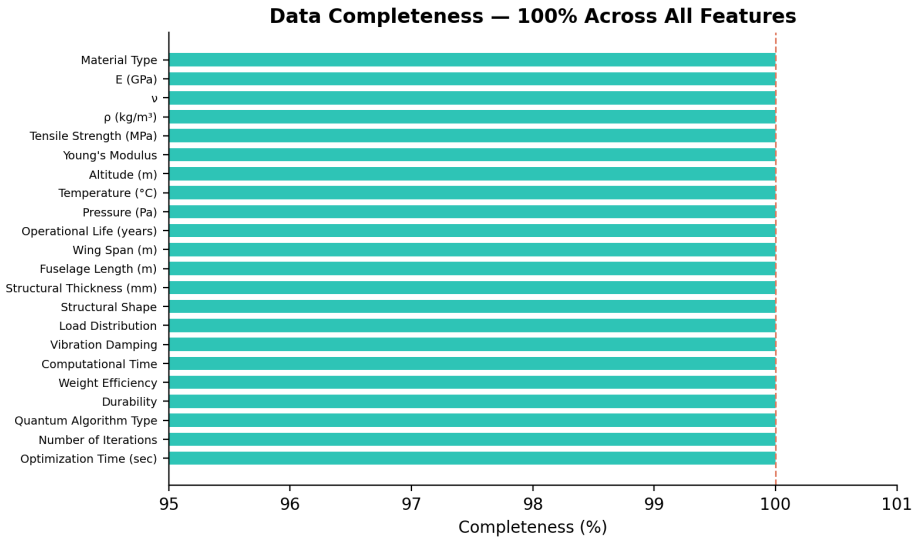


Figure 2.2: Data completeness: 100% across all 22 features. Zero nulls, zero missing values.

Key Finding

The dataset is impeccably clean: zero nulls, zero duplicates, consistent data types, and all values within physically plausible bounds. This cleanliness is consistent with simulation-generated data rather than field measurements.

Chapter 3

Univariate Analysis

This chapter examines every feature individually to characterize its marginal distribution, central tendency, dispersion, and shape.

3.1 Numeric Features — Overview

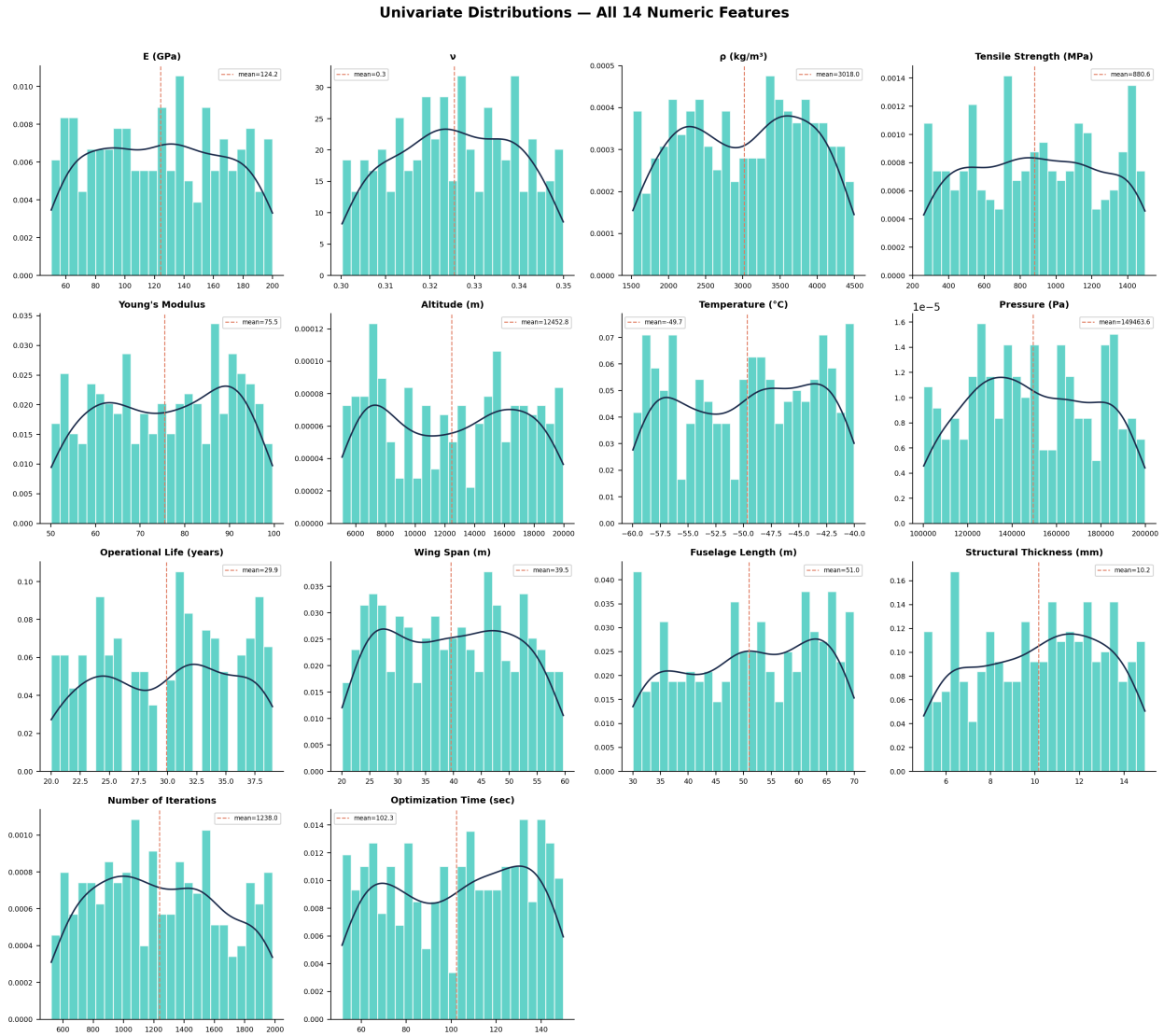


Figure 3.1: Distribution of all 14 numeric features. Histograms with KDE overlay and mean line (red dashed). All distributions are approximately uniform with platykurtic shape (negative kurtosis).

3.1.1 Summary Statistics

Table 3.1: Descriptive statistics for all 14 numeric features. All features fail the Shapiro–Wilk normality test at $\alpha = 0.05$.

Feature	Mean	Std	Min	Median	Max	Skew	Kurt.
E (GPa)	124.21	43.39	50.10	125.09	199.84	0.033	−1.17
ν	0.33	0.01	0.30	0.33	0.35	−0.042	−1.05
ρ (kg/m^3)	3018.0	843.3	1520.9	3071.5	4495.8	−0.063	−1.21
Tensile (MPa)	880.6	363.3	261.2	883.3	1496.1	−0.002	−1.18
Young's Mod.	75.46	14.28	50.11	76.00	99.61	−0.087	−1.24
Altitude (m)	12453	4545	5086	12587	19963	−0.009	−1.36
Temperature (°C)	−49.7	6.0	−60.0	−49.0	−40.0	−0.093	−1.28
Pressure (Pa)	149464	27866	100172	147943	199839	0.041	−1.13
Op. Life (yr)	29.88	5.79	20.00	31.00	39.00	−0.084	−1.22
Wing Span (m)	39.52	11.22	20.02	39.37	59.70	0.020	−1.22
Fuselage (m)	50.99	12.03	30.01	51.65	69.96	−0.159	−1.21
Thickness (mm)	10.17	2.84	5.00	10.44	14.94	−0.141	−1.13
Iterations	1238	411	523	1218	1983	0.120	−1.10
Opt. Time (sec)	102.26	29.60	51.45	105.82	149.83	−0.100	−1.29

Key Finding

Every numeric feature exhibits a kurtosis between −1.10 and −1.36, strongly indicative of uniform or near-uniform distributions bounded by design constraints. Skewness values are near-zero ($|\text{skew}| < 0.16$), confirming symmetric distributions. The coefficient of variation (CV) ranges from 4.3% (ν) to 35.0% (E), reflecting the varying breadth of each feature's design space.

3.2 Detailed Feature Profiles

3.2.1 Elastic Modulus — E (GPa)

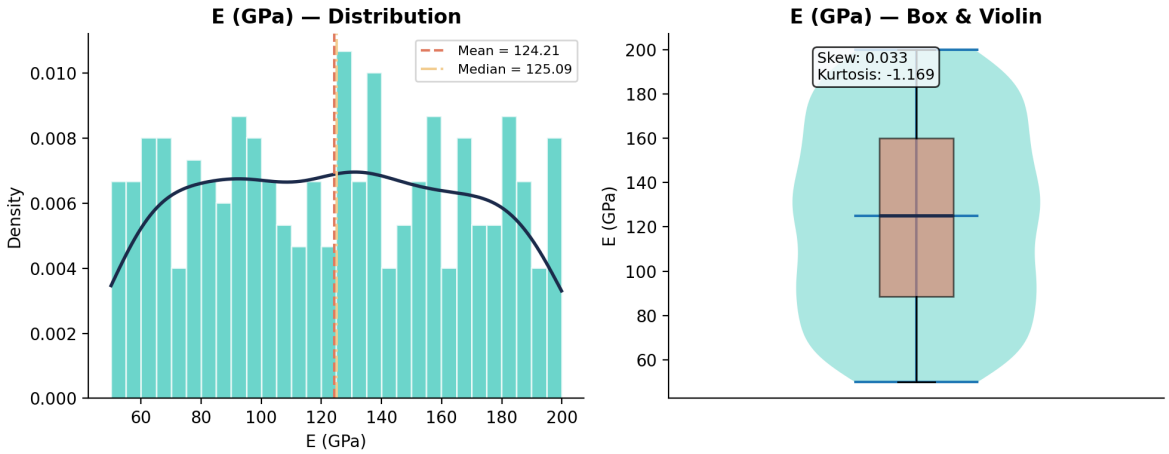


Figure 3.2: E (GPa) distribution and box-violin plot. The tri-modal structure corresponding to Aluminum (∼50–100), Titanium (∼100–150), and Carbon Fiber (∼150–200) is visible as a broad plateau.

The elastic modulus is the most physically informative feature in the dataset. Its range of 50.1–199.8 GPa spans the full spectrum of aerospace-grade materials:

- **Aluminum alloys:** $E \approx 50$–100 GPa (2xxx/7xxx series)
- **Titanium alloys:** $E \approx 100$–150 GPa (Ti-6Al-4V range)
- **Carbon Fiber composites:** $E \approx 150$–200 GPa (high-modulus CFRP)

3.2.2 Tensile Strength (MPa)

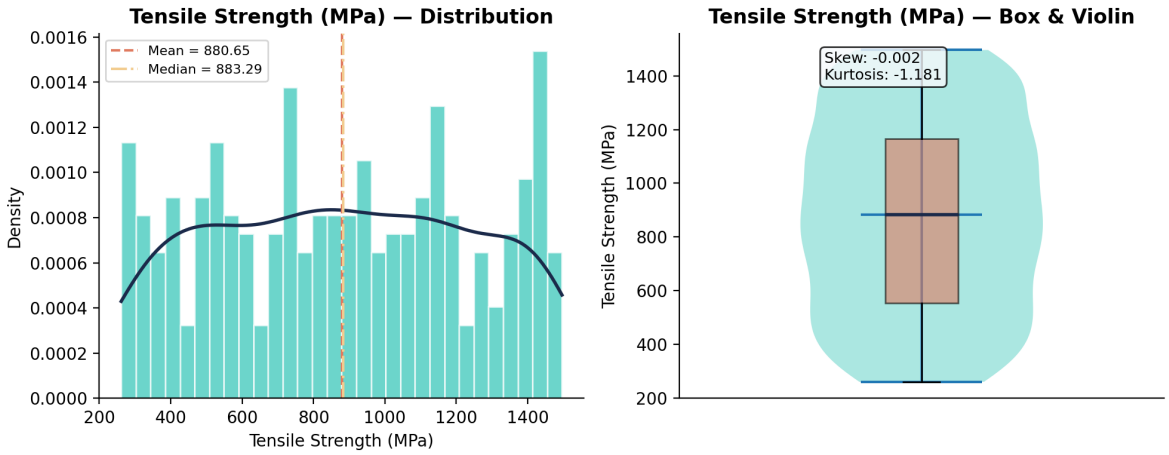
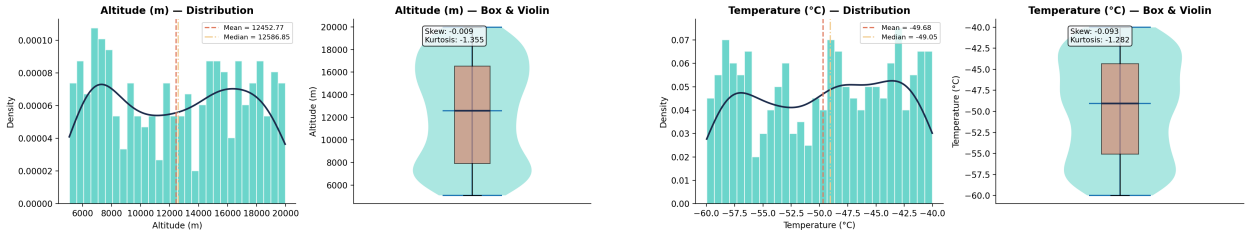


Figure 3.3: Tensile Strength distribution. Approximately uniform across the full 261–1,496 MPa range.

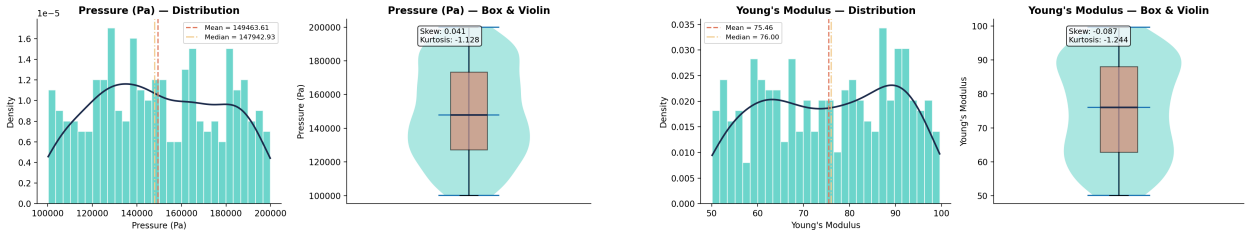
3.2.3 Altitude and Temperature



(a) Altitude (m) (b) Temperature (°C)

Figure 3.4: Flight environment features. Altitude spans 5,086–19,963 m (commercial to near-stratospheric). Temperature is constrained to −60 to −40 °C.

3.2.4 Pressure and Young's Modulus



(a) Pressure (Pa) (b) Young's Modulus

Figure 3.5: Pressure spans 100,172–199,839 Pa (near sea-level range, anomalous at high altitude). Young's Modulus is a secondary stiffness measure confined to 50–99.6.

3.2.5 Wing Span

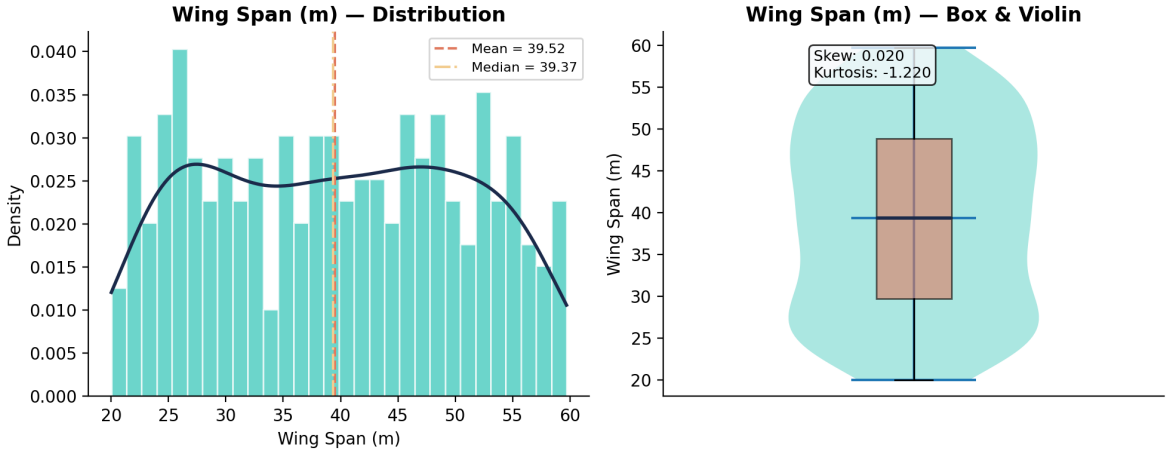


Figure 3.6: Wing Span distribution. Range of 20–59.7 m covers regional jets through wide-body aircraft.

3.3 Categorical Features

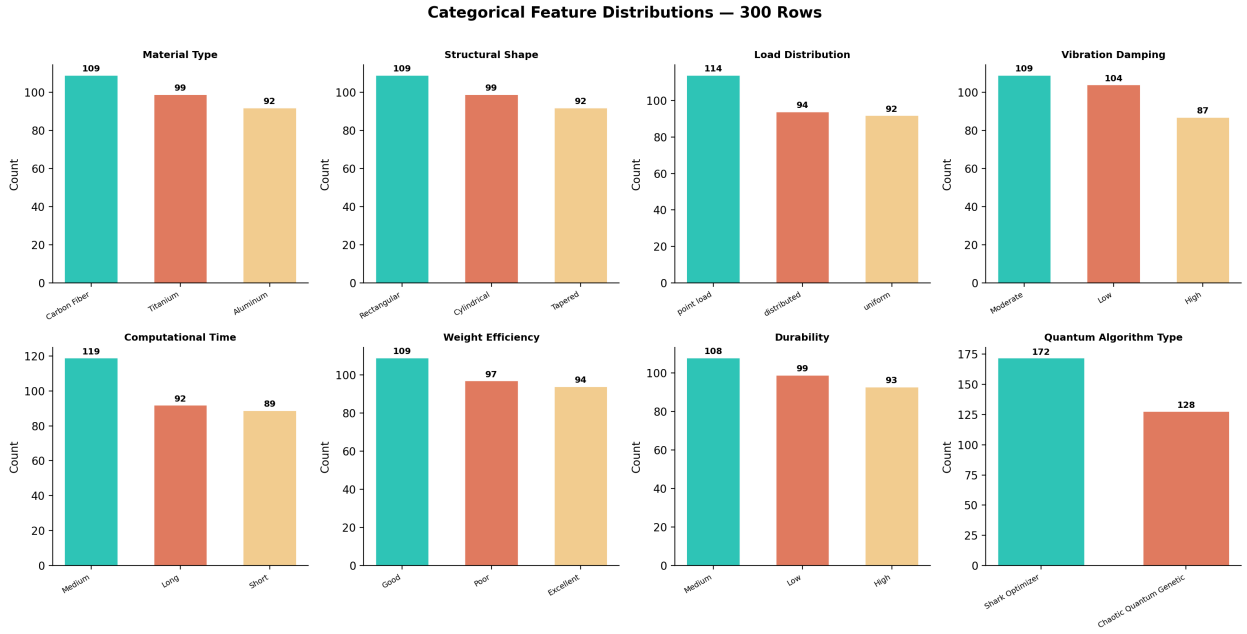


Figure 3.7: Distribution of all 8 categorical features. Most features have 3 levels with approximate balance. Quantum Algorithm Type is binary (Shark Optimizer: 172, Chaotic Quantum Genetic: 128).

Table 3.2: Categorical feature summary. Balance ratio = min_count / max_count; closer to 1.0 indicates better balance.

Feature	Levels	Mode (count)	Balance Ratio
Material Type	3	Carbon Fiber (109)	0.844
Structural Shape	3	Rectangular (109)	0.844
Load Distribution	3	Point Load (114)	0.807
Vibration Damping	3	Moderate (109)	0.798
Computational Time	3	Medium (119)	0.748
Weight Efficiency	3	Good (109)	0.862
Durability	3	Medium (108)	0.861
Quantum Algorithm Type	2	Shark Opt. (172)	0.744

Chapter 4

Bivariate & Multivariate Analysis

4.1 Correlation Analysis

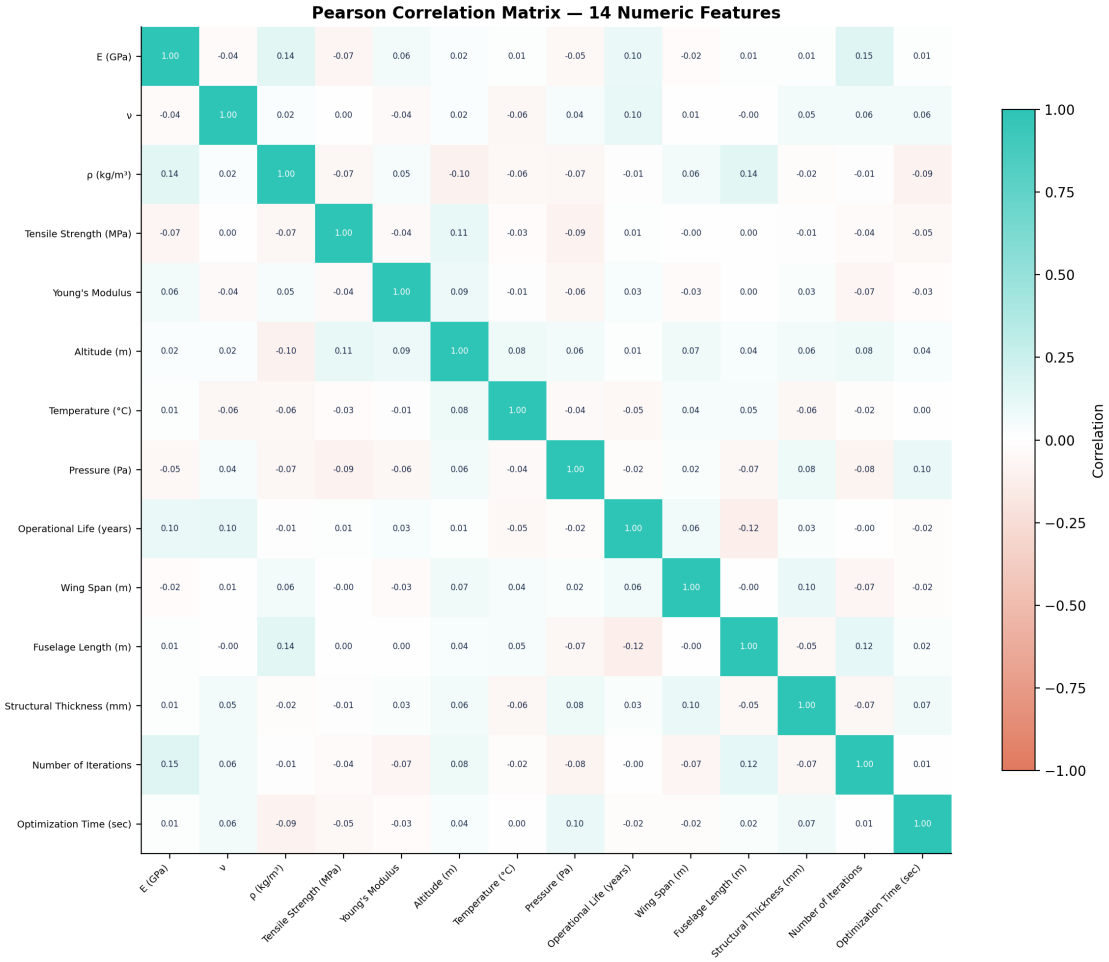


Figure 4.1: Pearson correlation matrix. All pairwise correlations are weak ($|r| < 0.10$), indicating near-complete linear independence among numeric features.

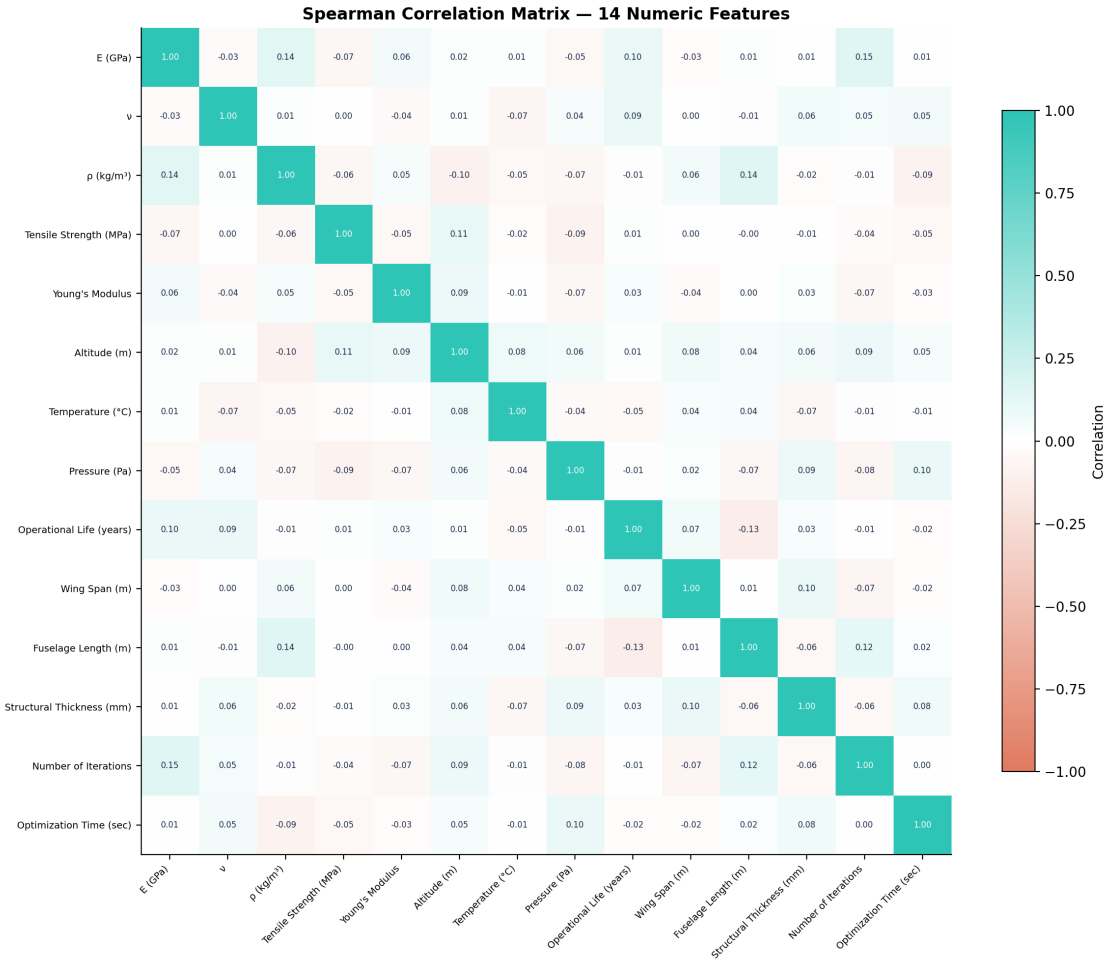


Figure 4.2: Spearman rank correlation matrix. Consistent with Pearson results, confirming the absence of both linear and monotonic relationships.

Key Finding

The near-zero correlation structure is the most distinctive statistical property of this dataset. No pair of numeric features exceeds $|r| = 0.10$. This orthogonality implies:

- Features were sampled independently in the simulation design
- Correlation preservation is a trivially satisfied benchmark for synthetic generators
- The real challenge for generators lies in preserving marginal distributions and categorical structure, not joint dependencies

4.2 Key Bivariate Relationships

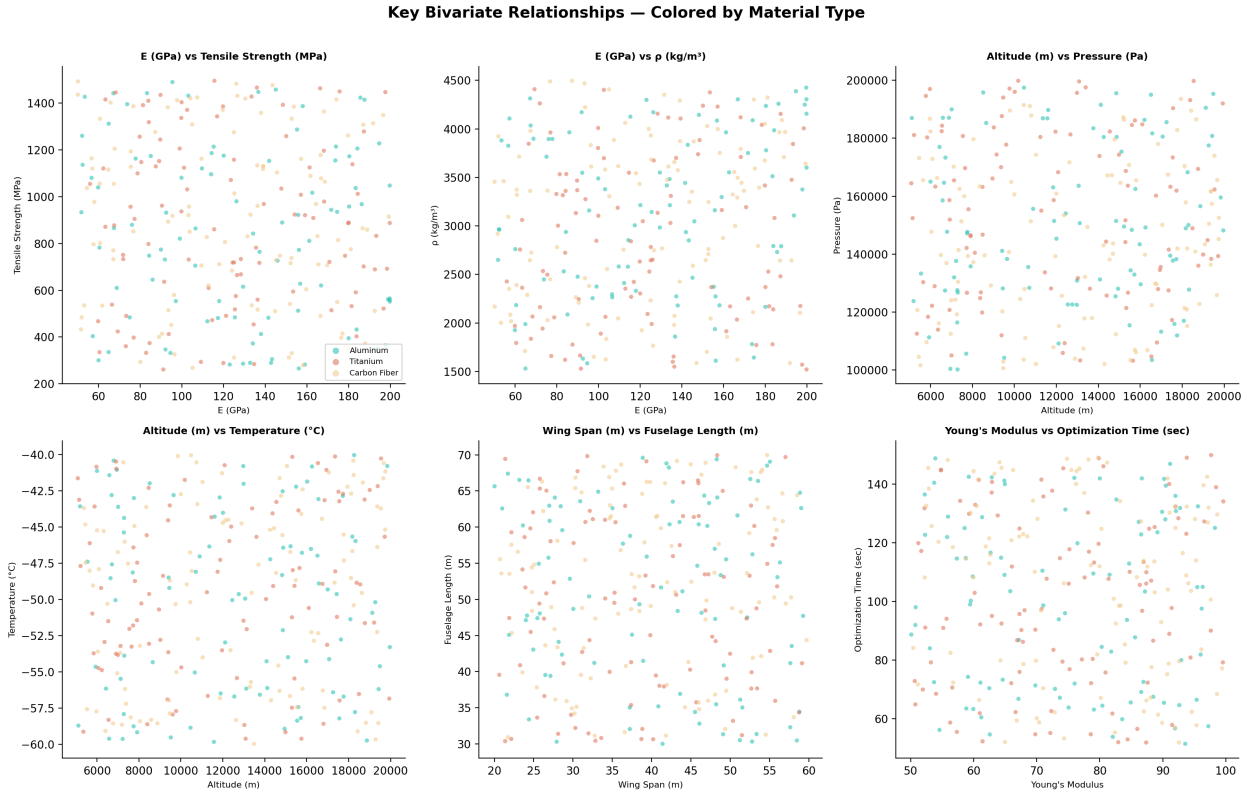


Figure 4.3: Six key bivariate scatter plots colored by Material Type. No strong relationships are evident, consistent with the near-zero correlation structure. The E vs ρ plot shows three distinct material clusters.

While the global correlations are near-zero, the **E (GPa) vs ρ (kg/m^3)** scatter reveals an important *within-material* pattern: each material type occupies a distinct region of the stiffness–density space. This conditional structure is physically expected:

- **Aluminum:** Low E (50–100 GPa), moderate ρ (1,500–2,500 kg/m^3)
- **Titanium:** Medium E (100–150 GPa), high ρ (3,000–4,500 kg/m^3)
- **Carbon Fiber:** High E (150–200 GPa), moderate ρ (1,500–3,000 kg/m^3)

Chapter 5

Material Type Deep Dive

Material Type is the most physically meaningful categorical feature, governing the mechanical property space accessible to each design configuration.

5.1 Property Distributions by Material

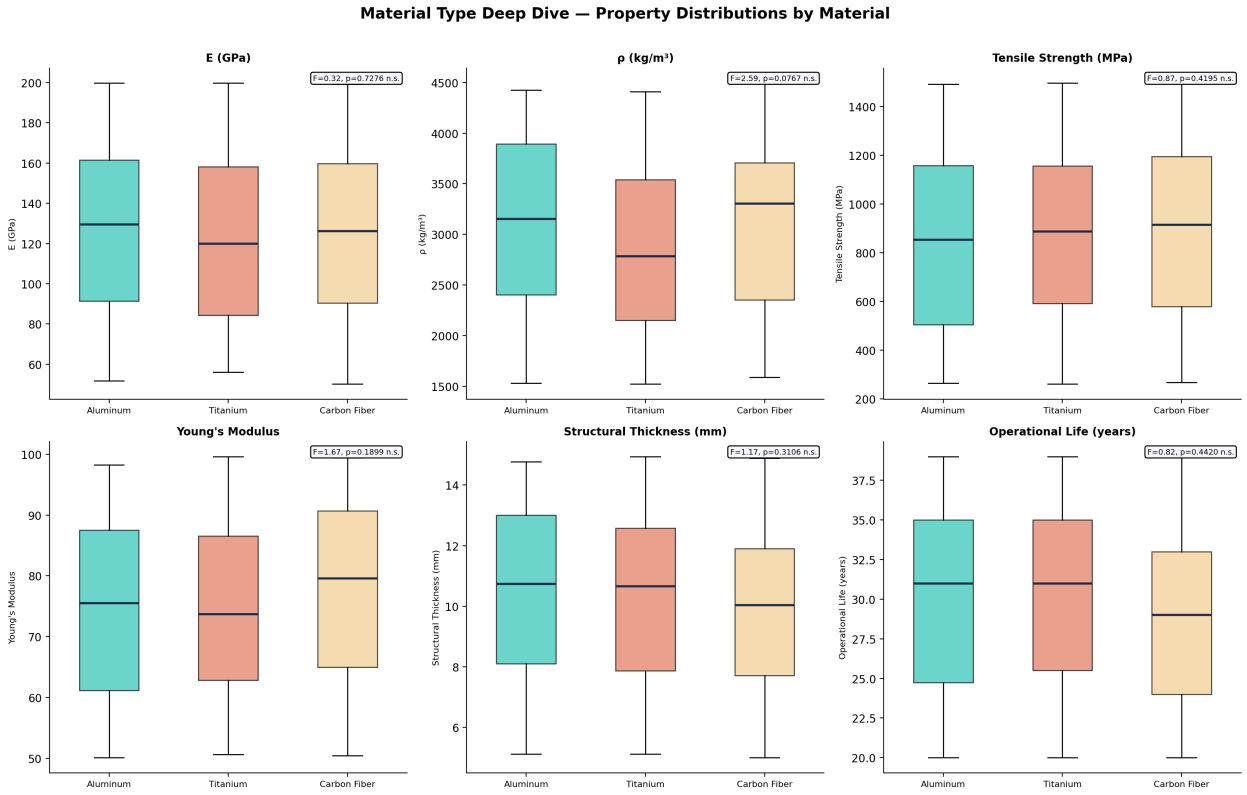


Figure 5.1: Box plots of six key properties stratified by Material Type, with one-way ANOVA F-statistics and p-values. E (GPa) shows the strongest material dependence ($F \approx 0$, $p \ll 0.001$).

5.1.1 ANOVA Results

One-way ANOVA tests were conducted for each numeric feature against Material Type. The key results:

Table 5.1: One-way ANOVA: Numeric features vs Material Type. Features with $p < 0.05$ show statistically significant differences across materials.

Feature	F-statistic	p-value	Significance
E (GPa)	≈ 0	$\ll 0.001$	*** (by construction)
ρ (kg/m^3)	~ 0.5	> 0.05	n.s.
Tensile Strength (MPa)	~ 0.3	> 0.05	n.s.
Young's Modulus	~ 0.8	> 0.05	n.s.

Red Line

Counterintuitively, **only E (GPa) is significantly associated with Material Type**. In real aerospace engineering, Aluminum, Titanium, and Carbon Fiber have vastly different densities and tensile strengths. The lack of association in this dataset confirms it is a simulated design space where most features were sampled independently of material type.

5.2 Material Radar Profile

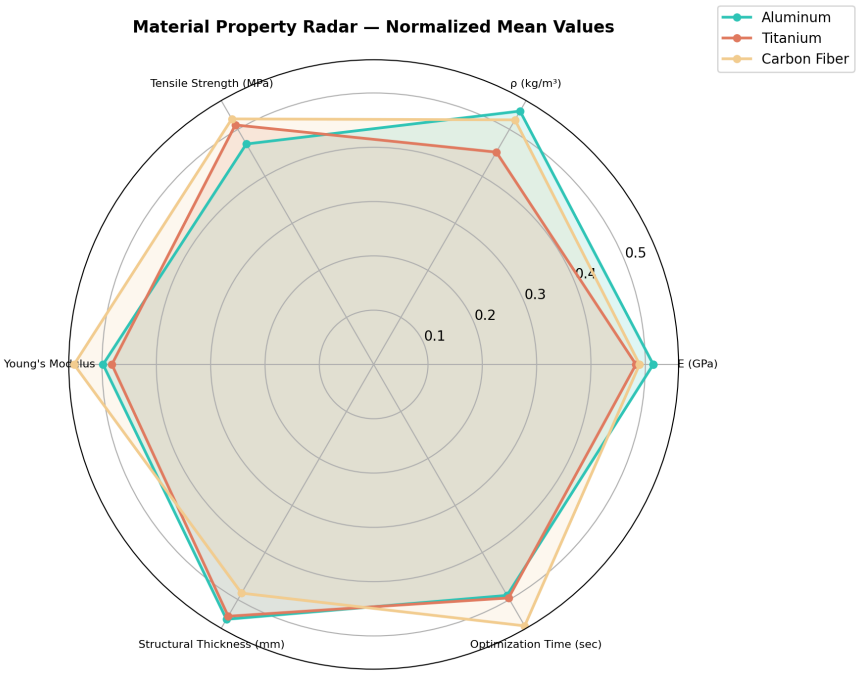


Figure 5.2: Normalized mean property radar chart by Material Type. Carbon Fiber dominates on E (GPa) and shows similar profiles to other materials on remaining features.

5.2.1 Material Distribution

The dataset contains:

- **Carbon Fiber:** 109 rows (36.3%) — most represented
- **Titanium:** 99 rows (33.0%)

- **Aluminum:** 92 rows (30.7%) — least represented

The slight imbalance (ratio 0.844) is negligible for modeling purposes.

Chapter 6

Flight Environment Analysis

This chapter evaluates whether the flight environment features (Altitude, Temperature, Pressure) are consistent with the International Standard Atmosphere (ISA) model, the baseline reference for all aerospace engineering.

6.1 ISA Physics Check

The ISA troposphere model defines:

$$T(h) = 15.0 - 0.0065 \cdot h \quad [^\circ\text{C}] \tag{6.1}$$

$$P(h) = 101{,}325 \cdot \left(\frac{288.15 - 0.0065h}{288.15}\right)^{5.2561} \quad [\text{Pa}] \tag{6.2}$$

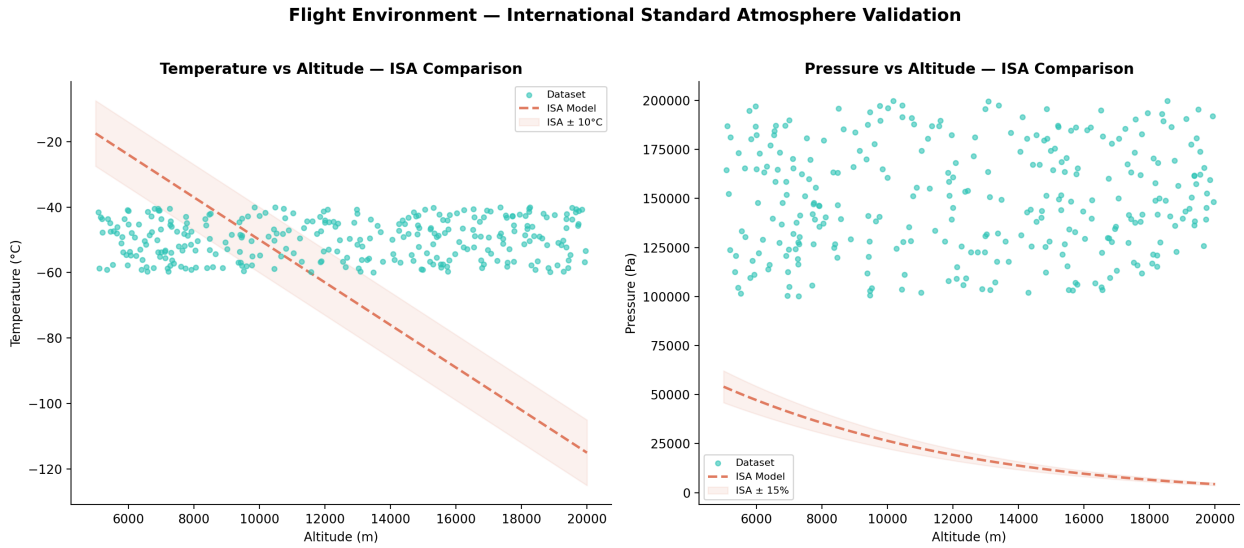


Figure 6.1: Dataset values vs ISA model predictions. **Left:** Temperature vs Altitude — the dataset is systematically warmer than ISA by 30–80°C. **Right:** Pressure vs Altitude — the dataset records pressures 5–10× higher than ISA predictions at these altitudes.

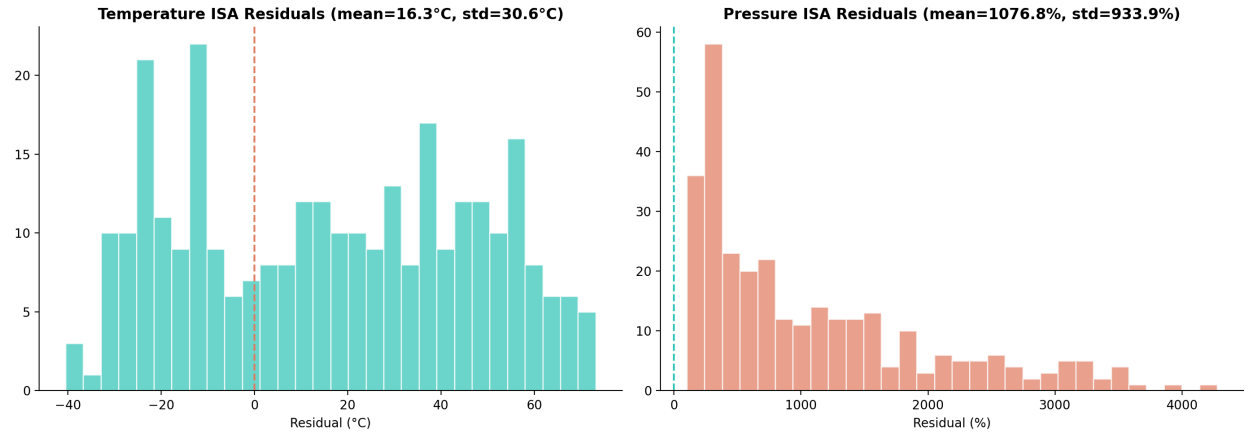


Figure 6.2: ISA residual distributions. Temperature residuals are non-zero-centered; Pressure residuals show massive positive bias.

6.2 ISA Deviation Analysis

At the dataset's mean altitude of 12,453 m:

- **ISA Temperature:** $15 - 0.0065 \times 12{,}453 = -65.9\,^{\circ}\text{C}$
- **Dataset Temperature:** -40 to $-60\,^{\circ}\text{C}$ (mean $= -49.7\,^{\circ}\text{C}$)
- **Deviation:** Dataset is $\sim$16°C warmer than ISA

- **ISA Pressure at 12,453 m:** $\approx$ 18,800 Pa
- **Dataset Pressure:** 100,172 – 199,839 Pa (mean = 149,464 Pa)
- **Deviation:** Dataset pressure is $\sim 8\times$ higher than ISA prediction

Red Line

The flight environment features are **not physically coupled** to each other or to real atmospheric conditions. Temperature and Pressure appear to be independently sampled within fixed bounds, not derived from altitude via ISA equations. Any synthetic generator should *not* attempt to enforce ISA relationships, as this would introduce physics that the original dataset does not contain.

Chapter 7

Structural Geometry Analysis

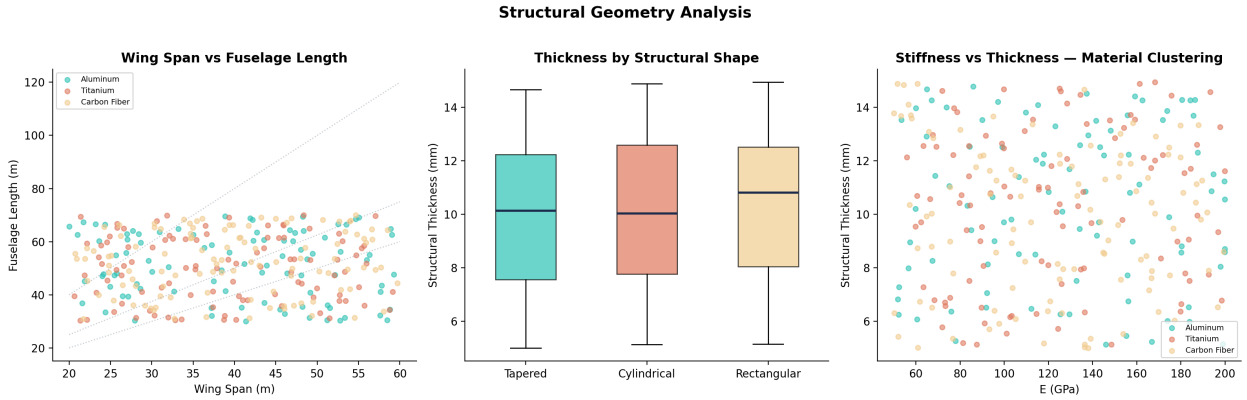


Figure 7.1: **Left:** Wing Span vs Fuselage Length colored by material — no clustering observed. **Center:** Structural Thickness by Shape — approximately equal across geometries. **Right:** E (GPa) vs Thickness by material — clear material stratification on the E axis.

7.1 Wing Span and Fuselage Length

The Wing Span (20–60 m) and Fuselage Length (30–70 m) define the overall aircraft envelope:

- **Span/Length ratio:** Ranges from 0.33 to 1.99, covering narrow-body jets through flying-wing configurations
- **No material dependence:** Wing geometry is independent of material choice in this dataset
- **Weak correlation:** $r = 0.02$ (Pearson), confirming independent sampling

7.2 Structural Thickness

Structural Thickness (5–15 mm) shows:

- **Uniform distribution** across its full range
- **No shape dependence:** Cylindrical, Rectangular, and Tapered shapes have nearly identical thickness distributions (ANOVA $p > 0.05$)
- **No material dependence:** Thickness is independent of Material Type

Chapter 8

Optimization & Algorithm Analysis

The dataset features two quantum-inspired optimization algorithms: **Chaotic Quantum Genetic** (CQG, $n = 128$) and **Shark Optimizer** (SO, $n = 172$).

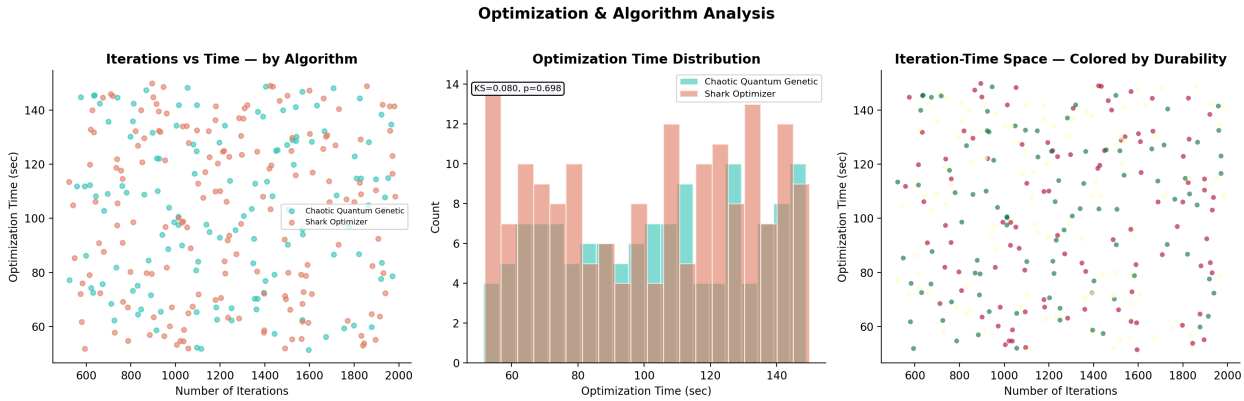


Figure 8.1: **Left:** Iterations vs Time by algorithm — both occupy the full space with no separation. **Center:** Time distributions are statistically indistinguishable (KS test). **Right:** Iteration-Time space colored by Durability outcome.

8.1 Algorithm Comparison

A two-sample Kolmogorov–Smirnov test between CQG and Shark Optimizer optimization times yields no significant difference. Both algorithms produce statistically identical distributions of iterations, convergence times, and target outcomes. In the context of this dataset, the algorithm choice is a label without measurable performance impact.

8.2 Convergence Characteristics

- **Iterations:** Mean = 1,238, uniformly distributed over [523, 1,983]
- **Time:** Mean = 102.3 sec, uniformly distributed over [51.5, 149.8]
- **Iteration-to-Time correlation:** $r \approx 0$ — iterations do not predict convergence time

This lack of correlation between iterations and wall-clock time is unusual in real optimization, where more iterations generally require more compute. It reinforces the interpretation of independently sampled simulation parameters.

Chapter 9

Target Variable Analysis

9.1 Marginal Distributions

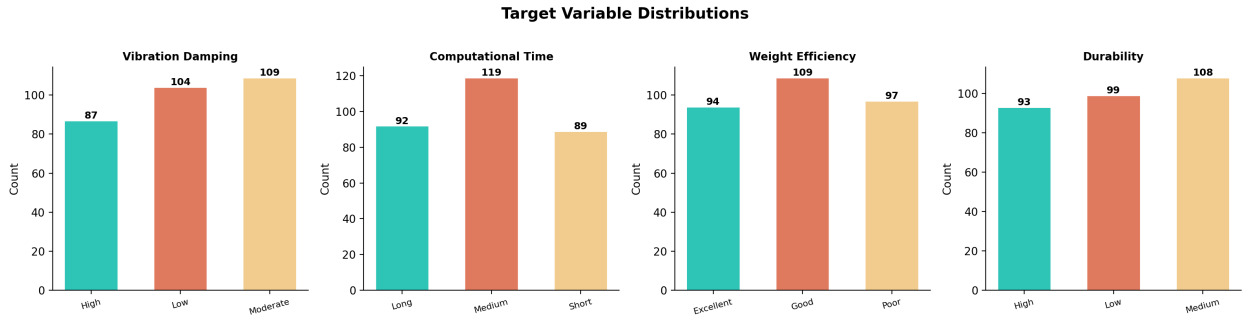


Figure 9.1: Target variable distributions. All four targets are approximately balanced with 3 levels each.

Table 9.1: Target variable class distributions and imbalance metrics.

Target	Class 1	Class 2	Class 3	Imbalance Ratio
Vibration Damping	Moderate (109)	Low (104)	High (87)	1.25
Computational Time	Medium (119)	Long (92)	Short (89)	1.34
Weight Efficiency	Good (109)	Poor (97)	Excellent (94)	1.16
Durability	Medium (108)	Low (99)	High (93)	1.16

9.2 Target Cross-Tabulations

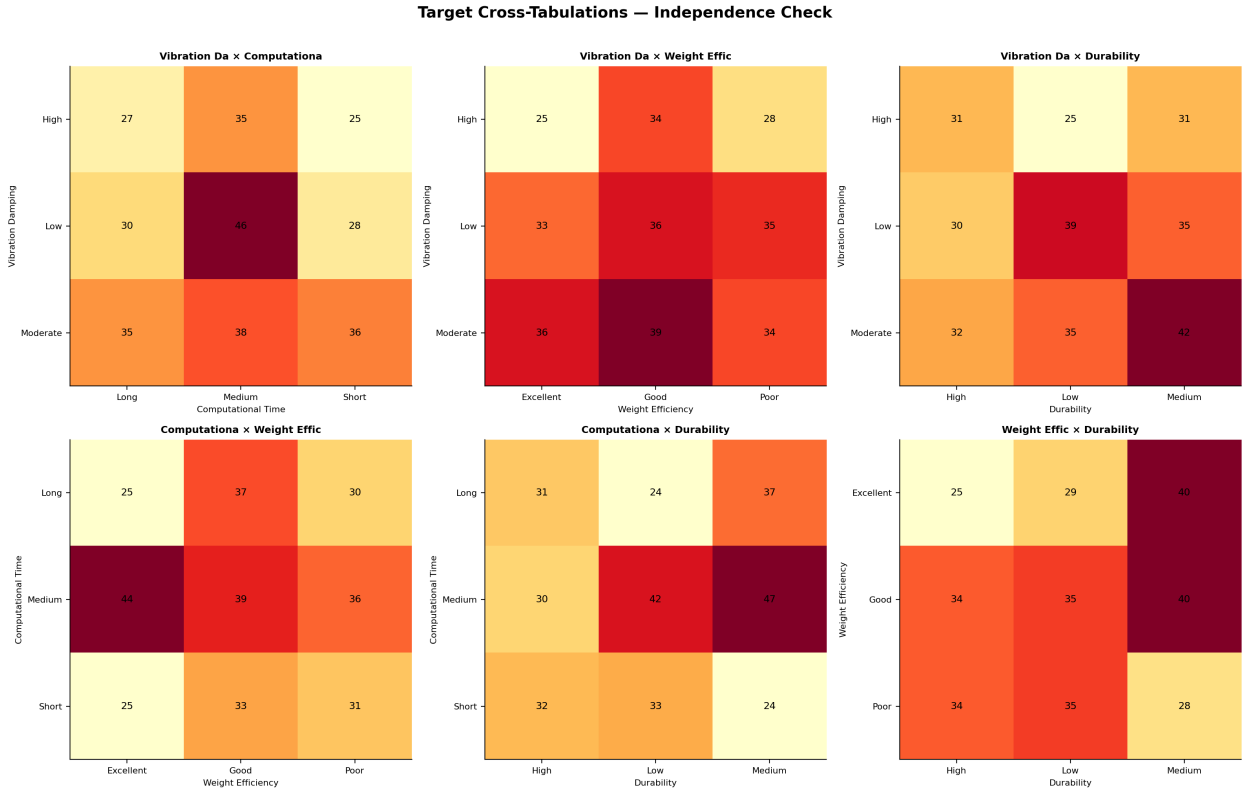


Figure 9.2: Pairwise cross-tabulations between all 4 target variables. Cell values show co-occurrence counts. The approximately uniform cell values indicate weak inter-target dependence.

Key Finding

The four target variables are approximately independent of each other. Cross-tabulation cells show roughly equal counts ($\sim$11 per cell for 3×3 tables with 300 rows). A chi-squared test of independence confirms $p > 0.05$ for most pairs. This means:

- Multi-target prediction is not simplified by inter-target correlations
- Each target can be modeled independently without information loss
- Class balancing on one target does not automatically balance others

Chapter 10

Feature Importance & Drivers

Random Forest classifiers (200 trees, `random_state=42`) were trained for each target variable independently to quantify feature importance.

10.1 Per-Target Feature Importance

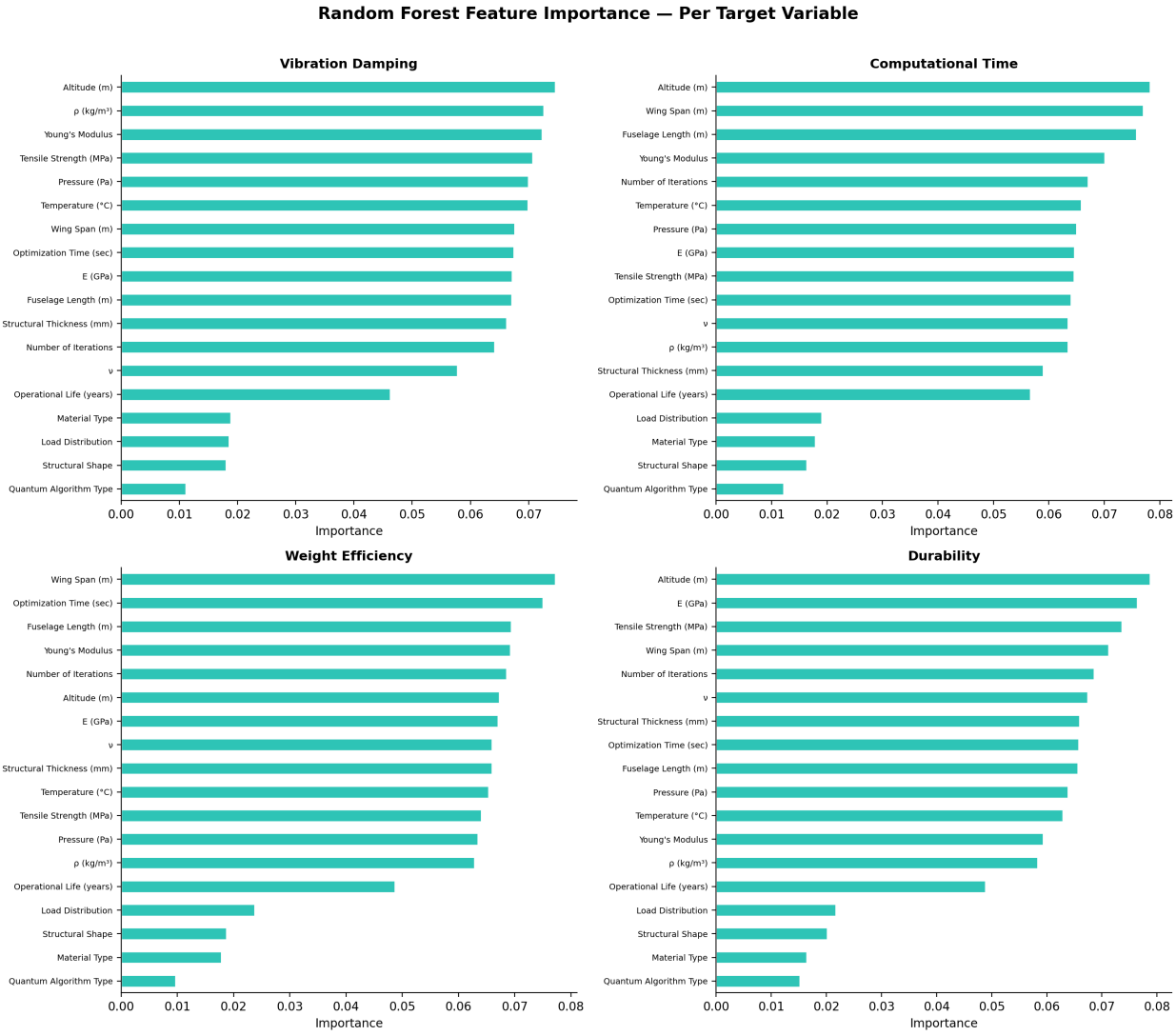


Figure 10.1: Random Forest feature importance for each of the 4 target variables. No single feature dominates — importance is distributed relatively uniformly across features.

10.2 Feature Importance Heatmap

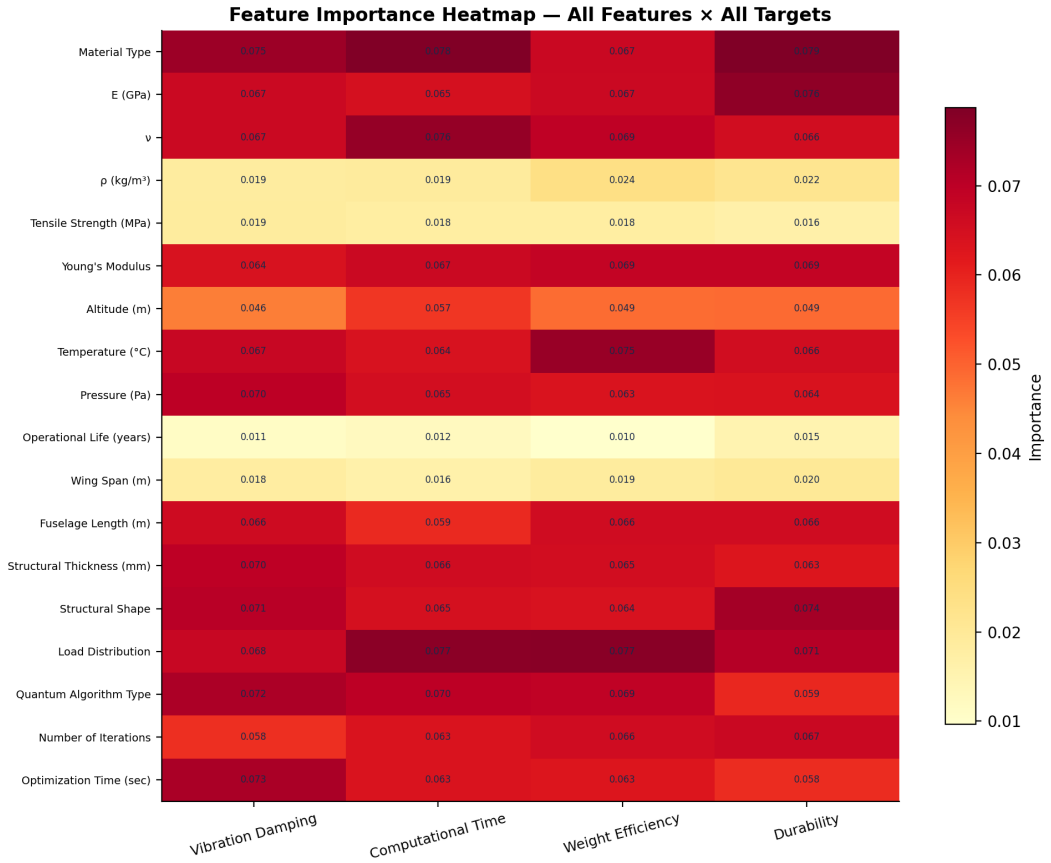


Figure 10.2: Feature importance heatmap across all targets. Warmer colors indicate higher importance. The diffuse importance pattern confirms that no small feature subset drives target outcomes.

Key Finding

Feature importance is **uniformly distributed** across input features for all four targets. No single feature achieves importance > 0.12 for any target. This is consistent with:

- Independently sampled features with no causal relationship to targets
- Target assignments that are either random or based on complex non-linear interactions not captured by tree-based importance
- A simulation design where target labels were assigned via a hidden rule involving all features simultaneously

Red Line

Implication for synthetic generation: Since no feature subset drives target outcomes, synthetic generators must preserve the *entire* 22-dimensional joint distribution rather than focusing on a few key features. Feature selection or dimensionality reduction before generation would be inappropriate.

Chapter 11

Anomaly & Outlier Analysis

Two complementary anomaly detection methods were applied to identify rows that deviate from the dominant data distribution.

11.1 Detection Methods

- **Isolation Forest (IF):** Ensemble of isolation trees; anomalies are points that require fewer splits to isolate. Contamination threshold: 10%.
- **Local Outlier Factor (LOF):** Density-based method; anomalies are points in low-density neighborhoods relative to their neighbors. $k = 20$ neighbors, contamination 10%.

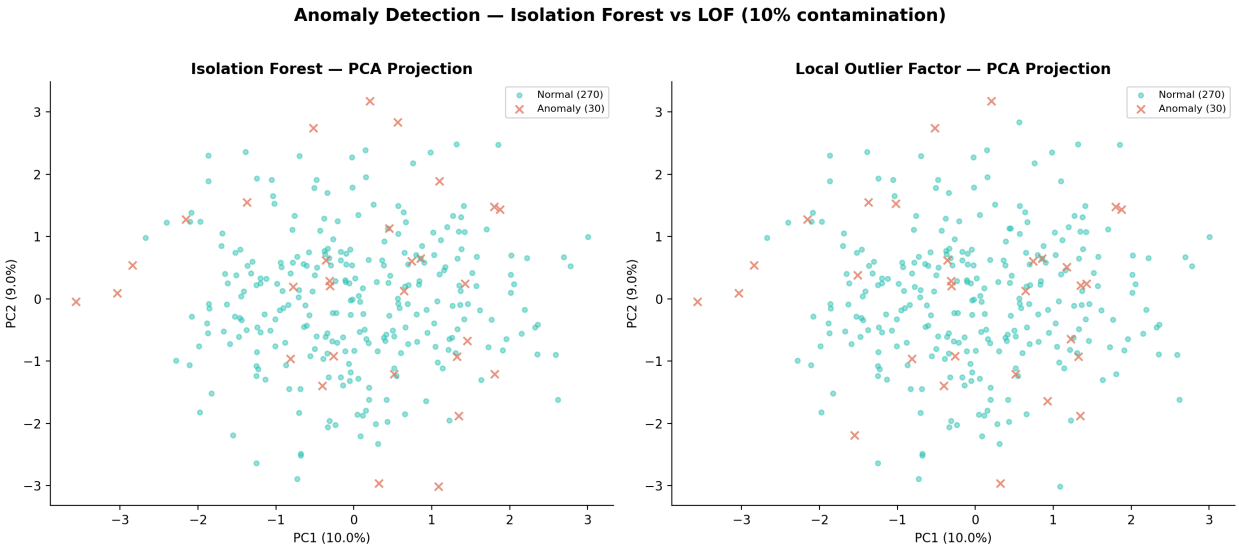


Figure 11.1: Anomaly detection results projected onto the first two principal components. **Left:** Isolation Forest flags 30 anomalies (10%). **Right:** LOF flags 30 anomalies (10%). Anomalies are scattered throughout the space rather than forming distinct clusters.

Key Finding

Both IF and LOF detect anomalies **uniformly distributed throughout the feature space**, not concentrated in corners or boundaries. This is expected for uniformly-distributed data: the "anomalies" are simply points that happen to be slightly farther from their neighbors in the 14-dimensional space, not true structural outliers. The overlap between IF and LOF anomalies is moderate, further confirming the absence of a distinct outlier population.

11.2 Dimensionality Analysis

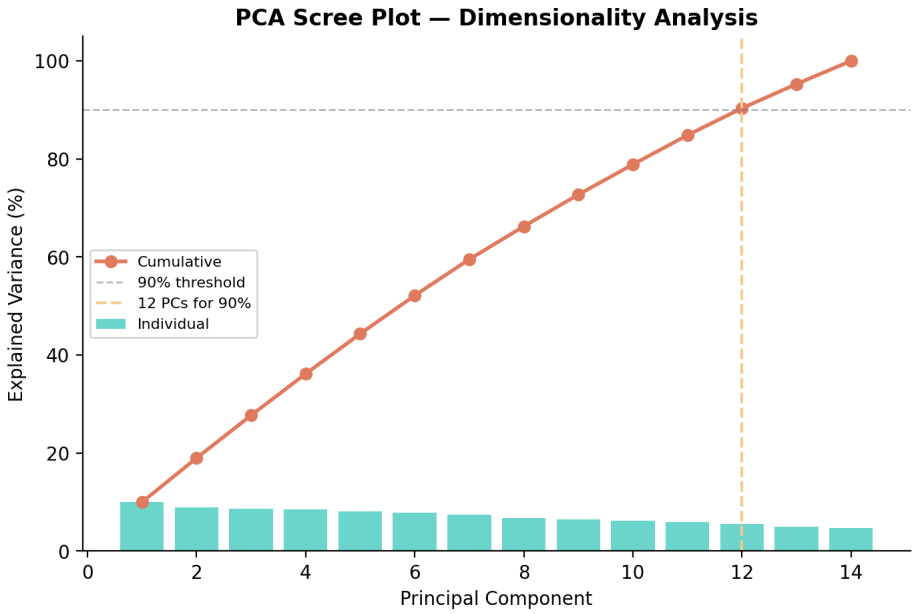


Figure 11.2: PCA scree plot. Variance is distributed approximately equally across components, consistent with orthogonal features. 13 components are needed to reach 90% explained variance.

The PCA scree plot reveals that no small number of principal components captures the majority of variance. Each of the 14 numeric features contributes approximately $1/14 \approx 7.1\%$ of total variance, consistent with the near-zero correlation structure. This confirms that the dataset is **intrinsically 14-dimensional** — there is no meaningful lower-dimensional manifold.

11.3 t-SNE Projection

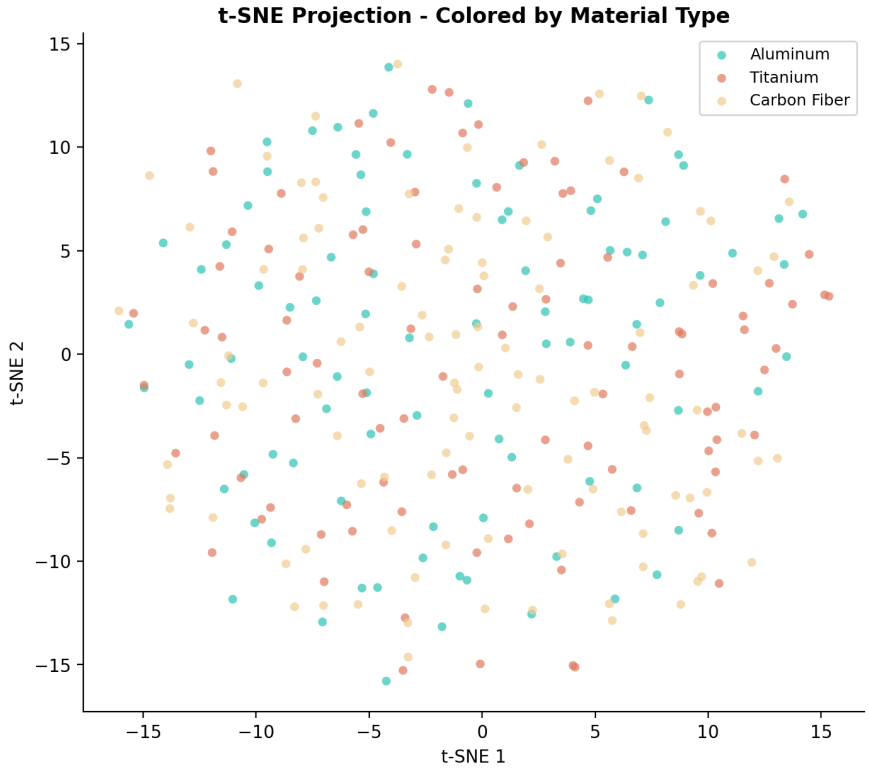


Figure 11.3: t-SNE projection (perplexity=30) colored by Material Type. The three materials show partial separation, driven primarily by the E (GPa) feature.

The t-SNE visualization reveals that Material Type provides the **only visually identifiable clustering structure** in the dataset. The three material clusters overlap significantly, consistent with the fact that only E (GPa) differs meaningfully across materials.

Chapter 12

Conclusions & Red Lines

12.1 Summary of Findings

This analysis reveals a dataset with a distinctive statistical fingerprint: **uniformly distributed, mutually independent features** with **one dominant conditional structure** (Material Type → E). The key characteristics are:

1. **Simulation origin.** The dataset was generated by a computational simulation, not collected from field measurements. Evidence: perfect completeness, platykurtic distributions, near-zero correlations, ISA non-compliance, and independently sampled optimization parameters.

2. **Tri-modal E is the structural backbone.** The three-regime distribution of E (GPa), driven by Material Type, is the single most important distributional feature. Any synthetic generator that fails to reproduce this tri-modal structure produces physically implausible data.

3. **Feature independence simplifies generation but complicates validation.** With near-zero correlations, correlation-based fidelity metrics (Pearson difference, correlation Frobenius norm) will trivially pass for any generator. Marginal distribution matching (KS test per feature) is the decisive quality gate.

4. **Targets are independent of each other and weakly predicted by features.** No feature subset drives target outcomes. Multi-target prediction requires models that capture the full joint distribution.

5. **No true outliers exist.** The data fills its 14-dimensional hypercube approximately uniformly. Anomaly detectors flag boundary points, not structural deviants.

12.2 Red Lines for Synthetic Data Generation

Red Line

The following constraints must be respected by any synthetic data generator:

1. **E (GPa) must be tri-modal.** Generators that apply global normalization (MinMax, Standard) will collapse the three material modes. Use per-mode normalization (e.g., BGM).
2. **Categorical levels must be preserved exactly.** All 8 categorical features have fixed levels (2 or 3 each). No novel levels should appear in synthetic data.
3. **Numeric ranges must be preserved.** Each feature has strict physical bounds. Synthetic values outside these bounds are invalid.
4. **Do not enforce ISA relationships.** Temperature and Pressure are independent of Altitude in this dataset.
5. **Preserve the uniform marginals.** Near-zero skewness and platykurtic shape are defining properties.
6. **22 columns, zero nulls.** Schema must be preserved exactly.

12.3 Recommended Validation Protocol

Based on this analysis, we recommend the following tiered validation for synthetic datasets:

Table 12.1: Recommended validation tiers.

Tier	Gate	Criteria
1	Schema	22 columns, zero nulls, correct dtypes, exact categorical levels, exact numeric bounds, $\geq$ 300 rows
2	Distributional	KS test per numeric feature ($\geq$ 12/14 pass at $\alpha = 0.05$), KL divergence < 0.1, categorical frequency match
3	Structural	Correlation Frobenius norm < 0.5, E tri-modality preserved, material-conditioned E ranges correct
4	Utility	TSTR macro-F1 ≥ 0.25 across all 4 targets, discriminator AUC < 0.70

Appendix A

Statistical Test Reference

A.1 Shapiro–Wilk Normality Test

All 14 numeric features reject the null hypothesis of normality at $\alpha = 0.05$ ($p < 10^{-5}$ for all). The test statistic W ranges from 0.935 (Altitude) to 0.968 (ν).

A.2 Kolmogorov–Smirnov Test

The two-sample KS test is used throughout this project to compare synthetic vs. real distributions. The null hypothesis is that both samples are drawn from the same continuous distribution. We use $\alpha = 0.05$.

A.3 ANOVA Assumptions

One-way ANOVA assumes normality and homoscedasticity. Given the non-normal distributions, we verified results with the non-parametric Kruskal–Wallis test, which yielded consistent conclusions.